

Open multi-source inputs

H3 assembly (R9)

Learning & blocked evaluation

Diagnostics & outputs

Open flood labels
(DEP stormwater, 311, USGS Ida HWM)

Static predictors
(elevation, slope, impervious,
building density, distance-to-water)

Rainfall condition r
(constant synthetic rainfall;
not observed radar)

Join layers to H3 cells

Provenance tags:
assembly_mode · feature_source
label_source · rainfall_source

Gradient-boosting classifier
+ continuous risk regressor

H3-block GroupKFold spatial CV
(primary blocked evaluation)

Constant-class baselines
(always-positive & always-negative)

Baselines: logistic / ponding rule

$PFI_h(c,r)$ — rainfall-conditioned
cell probability/index (not PFIb;
definition/interface, scenario response flat)

Scale-loss Jaccard ladder
(R10 → R9 / R8)

Adaptive refinement
(PFI_h screens parents → R11)

Sandy negative-control check

Negative control
(FEMA Sandy — never a label)

